

Oral Health, Teeth and Periodontitis

Gingival disease and periodontitis (IL1A, IL1B, GLT6D1), caries, dental prostheses, aphthae, temporomandibular joint and jaws: polygenic scores and loci load

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

Contents

Section	Page
1. Summary	3
2. How to read this report	4
3. Polygenic scores	5
4. Key variants by topic	6
5. GWAS loci burden	7
6. Favourable	8
7. Unfavourable	9
8. Mechanisms of ageing	10
9. Practical suggestions	11
10. Limitations	12
11. Methods and sources	13
12. Annex: all variants analysed	14

1. Summary

Favourable. The "genotype IL-1 compound" of periodontitis is negative: Sérgio F has an IL1A –889T allele, but does not have the IL1B +3954T, and the classic test needs both. The combined score of gingival and periodontal diseases is at percentile 38.4. The loss of teeth / prostheses is at percentile 26.7 and the diseases of the jaws in the 47.7. The caries loci load and tooth loss is very low (2.7) and the aphthae load is also low (6.2).

Unfavourable. Genetic gingivitis is high (percentile 89.7). The periodontitis is typical-high (percentile 75.8; loci load at 70.9, with only 5 loci). It has G/G in the GLT6D1 rs1537415, the locus of aggressive periodontitis, with 2 alleles of risk (small effect). Temporomandibular joint dysfunction (ATM) is at percentile 87.4.

What does it mean? The genetic trend is for gums that inflame easily (gingivitis), but without the IL-1 profile of severe periodontitis. Gingivitis is reversible and is the gateway to periodontitis. Periodontitis binds to systemic inflammation, glucose and arteries. Therefore, the control of the plaque every day and regular periodontal surveillance are the measures with more return. Isotretinoin dry in the mouth and increases the risk of caries and gingivitis while taken.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 3 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 3 of the chip, 0 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 1 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 7 PGS Catalog scores calculated locally with 4,760,149 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 230 loci, 217 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In all these scores, a higher percentile = more risk (gingivitis, periodontitis, tooth loss, aphthae, ATM and jaws).

Gums and periodontium

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Periodontitis (acute or chronic)	PGS019313 (Chen 2025, PRS-CS, 984 thousand variants)	950,708 983,817 (weighted coverage 96%)	75.8	TYPICAL	typical
Gingivitis	PGS019308 (Chen 2025, PRS-CS, 984 thousand variants)	950,687 983,793 (weighted coverage 95%)	89.7	MORE LIKELY	unfavourable
Gingival and periodontal diseases	PGS019303 (Chen 2025, PRS-CS, 984 thousand variants)	950,710 983,820 (weighted coverage 94%)	38.4	TYPICAL	typical

Teeth and mucosa

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Dental prostheses (loss of teeth)	PGS000995 (Tanigawa 2022, 6042 variants)	5,319 6,042 (weighted coverage 89%)	26.7	TYPICAL	typical
Phects (ulcer of the mouth)	PGS000947 (Tanigawa 2022, 1648 variants)	1,436 1,647 (weighted coverage 88%)	77.4	TYPICAL	typical

Maxillary and ATM

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Temporomandibular joint dysfunction	PGS019323 (Chen 2025, PRS-CS, 984 thousand variants)	950,644 983,748 (weighted coverage 96%)	87.4	MORE LIKELY	unfavourable
Maxillary diseases	PGS019318 (Chen 2025, PRS-CS, 984 thousand variants)	950,645 983,750 (weighted coverage 96%)	47.7	TYPICAL	typical

4. Key variants by topic

Most studied candidate variants in periodontitis. The effect of each one is small; the composite IL-1 test (IL1A + IL1B) has mixed evidence.

Gingival inflammation and periodontitis

Interleukin-1 (IL-1 α and IL-1 β) controls the inflammatory response of the gums to the plaque and the resorption of the alveolar bone. GLT6D1 was the first GWAS locus of aggressive periodontitis.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
IL1A	rs1800587	-889C>T	G/A	chip	A (1 copy/s)	unfavourable (light)	-889C>T (A na strand +): the T allele in the gene increases IL-1 α ; it is part of the "compound IL-1 genotype" associated with more severe periodontitis (mixed evidence)	Weak
IL1B	rs1143634	+3954C>T	G/G	chip	A (0 copy/s)	favourable	+3954C>T (A na strand +): the T allele in the gene increases IL-1 β in the gingival fluid; the compound genotype IL1A+IL1B is associated with chronic periodontitis (mixed evidence)	Weak
GLT6D1	rs1537415		G/G	chip	G (2 copy/s)	unfavourable	The G allele is associated with aggressive periodontitis (first GWAS locus of periodontitis; it interferes with the GATA-3 binding)	Weak

5. GWAS loci burden

Sum of GWAS Catalog loci ($p \leq 510-8$) of caries, tooth loss and dentures, dental wear, periodontitis, gingival disease and aphthae. Alleles are oriented towards more risk and the sum is compared with the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Caries and tooth loss	133 (125)	73 of 250	124	2.7	favourable
Periodontitis and gingival disease	5 (5)	6 of 10	5	70.9	typical
Phects (ulcer of the mouth)	92 (87)	48 of 174	87	6.2	favourable

6. Favourable

- **IL-1 negative compound genotype:** IL1A –889 C/T (1 allele), but **IL1B +3954 C/C**. Without the pair associated with more severe chronic periodontitis and more bone loss. **Weak moderate.**
- **Gingival and periodontal diseases (combined score)** at percentile 38.4. **Weak moderate.**
- **Dental prostheses (loss of teeth)** at percentile 26.7. **Weak moderate.**
- **Caries and tooth loss:** load of loci in 2.7 (133 locus of caries, dentures and dental wear). **Moderate.**
- **Maxillary diseases** at percentile 47.7. **Weak.**
- **Phects:** low loci load (6.2), but the score is at percentile 77.4 (mixed signal, globally typical). **Weak.**

7. Unfavourable

- **Gingivitis** at percentile 89.7: gums tend to inflame and bleeding with less plaque. **Moderate.**
- **Periodontitis** at percentile 75.8; loci load at percentile 70.9 (only known 5 loci). **Weak moderate.**
- **GLT6D1 rs1537415 G/G**: 2 alleles of aggressive periodontitis (OR ~1.5 per allele in the original study; it is mainly relevant at young age). **Weak.**
- **IL1A -889 C/T**: 1 allele that increases IL-1 α (without IL1B, the effect is small). **Weak.**
- **ATM dysfunction** at percentile 87.4 (mandibular pain, clicks, bruxism). **Weak moderate.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Chronic Inflammation	Gingivitis at percentile 89.7; periodontitis is a continuous source of systemic inflammation	Unfavourable
Dysbiosis	Dysbiotic plaque (P. gingivalis and others) initiates periodontitis; the gums that inflame easily favor it	Unfavourable (mild)
Inflammaging	Chronic periodontitis increases systemic CRP and IL-6; the compound IL-1 genotype is negative	Mixed
Remodelling of the extracellular matrix	The destruction of the periodontal ligament and gingival collagen depends on metalloproteinases; combined score at percentile 38.4	Neutral
Osteoporosis	Loss of alveolar bone (loss of teeth at percentile 26.7)	Favourable (light)
Vascular Aging	Periodontitis is associated with atherosclerosis and cardiovascular events (via inflammation and bacteraemia)	Unfavourable (conditional)
Hyperglycemia and Intolerance to Glucose	The relationship is bidirectional: periodontitis worsens glucose and hyperglycemia worsens periodontitis	Unfavourable (conditional)

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Daily board control:** brushing 2×/day during 2 minutes with pulp of 1450 ppm of fluorine (rotational or sonic electric toothbrush) and interdental interdental brushes or wire every day. In high genetic gingivitis, cleaning between teeth is the most important step. *Suggestion (not applied).*
2. **Periodontal evaluation** (probing and recording of bleeding) with the dentist or hygienist, and professional hygiene every 6 months, or every 3–4 months if there are pockets ≥ 4 mm. *Suggestion (not applied).*
3. **Signals to monitor:** bleeding on brushing, swollen or retracted gums, persistent bad breath, teeth with mobility. Treating gingivitis early prevents periodontitis. *Suggestion (not applied).*
4. **Dry mouth by isotretinoin:** drinking water throughout the day, tablets or gel with xylitol, and paste or varnish with high concentration fluoride if there are new cavities. *Suggestion (not applied).*
5. **Avoid chlorhexidine mouthwashes for long periods** (only in course of 1–2 weeks, if the dentist indicates): stains, alteration of taste and destruction of beneficial oral bacteria. *Suggestion (not applied).*
6. **Sugar:** reducing the frequency (the number of times per day counts more than the amount) and not snacking between meals. *Suggestion (not applied).*
7. **Oral probiotic with Lactobacillus reuteri** (DSM 17938 + ATCC PTA 5289), in cycles of 4–12 weeks: reduces gingival bleeding in gingivitis and periodontitis trials. *Suggestion (not applied).*
8. **ATM (high percentile):** if there is pain in the mandible, click or bruxism, make a nocturnal drip made by the dentist and physiotherapy of ATM. Avoid chewing pellets for long periods. *Suggestion (not applied).*
9. **In the plan and useful for the gums:** Omega-3 3.2–4 g/day (periodontal anti-inflammatory), vitamin D of the plan, controlled glucose, zero tobacco and zero alcohol. *Suggestion (not applied).*
10. **Recurrent aphthae** (if they arise): request B12, folate and ferritin (maintaining ferritin below 100) and avoid pastes with sodium lauryl sulfate. *Suggestion (not applied).*

10. Limitations

- The compound IL-1 test has inconsistent replication and should not be used alone to decide the treatment.
- The GWAS of periodontitis found few loci (the heritability is low to moderate). The real state of the mouth depends mainly on plaque, tobacco and glucose.
- The scores of gingival disease, periodontitis, ATM and jaws (Chen 2025) are based on diagnoses of health records; gingivitis recorded in biobanks is underdiagnosed.
- The load of loci GWAS sums SNPs in linkage imbalance; it is descriptive.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the imputed VCF; in the palindromic SNPs the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog (7 scores: periodontitis, gingivitis, gingival and periodontal diseases, dental prostheses, aphthae, ATM and maxillary diseases). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 5 \times 10^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** reference literature (Kornman 1997 and Nikolopoulos 2008 for IL-1; Schaefer 2010 for GLT6D1; Shungin 2019 for caries and periodontitis; Chen 2025 and Tanigawa 2022 for scores; Twetman 2009 for *L. reuteri*), GWAS Catalog and ClinVar. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Gingival inflammation and periodontitis	IL1A	rs1800587	G/A	chip	A	1	strand +	1 copy of risk allele
Gingival inflammation and periodontitis	IL1B	rs1143634	G/G	chip	A	0	strand +	without the risk allele
Gingival inflammation and periodontitis	GLT6D1	rs1537415	G/G	chip	G	2	strand + (palindromic; assumed +)	2 risk allele copies